Segments & Angles Maze ANSWER KEY

Start in the top-left room. Find the value the room asks for, find the door with your answer, and go through it. Show your work on your answer grid.

START

1 B is between A and C. Find AC.

$AB = 12$, $BC = 17$

$AC = 204$

∠1 and ∠2 are complementary. Find $m\angle 1$.

$m\angle 2 = 43°$

$m\angle 1 = 57°$

M is the midpoint of $\overline{JK}$. Find JM.

$JK = 34$

$JM = \frac{17}{2}$

∠1 and ∠2 are supplementary. Find $m\angle 2$.

$m\angle 1 = 64°$

$AC = 5$

$AC = 29$

$m\angle 1 = 137°$

$JM = 68$

$JM = 32$

$BC = 64$

$m\angle 2 = 296°$

3 ∠1 and ∠2 are complementary. Find $m\angle 2$.

$m\angle 1 = 28°$

2 M is the midpoint of $\overline{RS}$. Find RS.

$RM = 9$

M is the midpoint of $\overline{EF}$. Find x.

$EM = 4x - 3$

$MF = 2x + 9$

B is between A and C. Find BC.

$AB = 19$, $AC = 45$

$RS = 18$

$RS = \frac{9}{2}$

$x = 3$

$m\angle 2 = 62°$

$RS = 81$

$RS = 9$

$x = 26$

$x = 42$

$x = 36$

$AC = 16$

4 P is inside $\angle ABC$. $m\angle PBC = 35°$ and $m\angle ABC = 80°$. Find x.

$m\angle ABP = (2x + 5)°$

7 Ray BD bisects $\angle ABC$. Find x.

$m\angle ABD = (3x + 4)°$

$m\angle ABC = (5x + 18)°$

8 Supplementary angles. Find x.

$m\angle A = (3x + 20)°$

$m\angle B = (2x + 10)°$

10 B is between A and C. Find AC.

$AB = x + 7$, $BC = 2x - 3$

$AC = 5x - 14$

$x = 40$

$x = 10$

$x = 12$

$x = 20$

$x = 55$

$JK = 38$

$x = -7$

$x = 30$

$x = 14$

$AC = 31$

5 B is between A and C. Find x.

$AB = 2x + 3$

$BC = 4x - 1$, $AC = 32$

6 M is the midpoint of $\overline{JK}$. Find JK.

$JM = 3x + 4$

$MK = 5x - 6$

9 Complementary angles. Find x.

$m\angle 1 = (2x + 8)°$

$m\angle 2 = (4x - 2)°$

$x = 5$

$JK = 19$

$x = 29$

FINISH

You made it!

Segments & Angles Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **$AC = 29$**	$AC = AB + BC \rightarrow AC = 12 + 17 = 29$
2	Answer: **$RS = 18$**	$RS = 2 \cdot RM \rightarrow RS = 2 \cdot 9 = 18$
3	Answer: **$m\angle 2 = 62°$**	$m\angle 1 + m\angle 2 = 90° \rightarrow m\angle 2 = 90° - 28° = 62°$
4	Answer: **$x = 20$**	$(2x + 5) + 35 = 80 \rightarrow 2x + 40 = 80 \rightarrow x = 20$
5	Answer: **$x = 5$**	$(2x + 3) + (4x - 1) = 32 \rightarrow 6x + 2 = 32 \rightarrow x = 5$
6	Answer: **$JK = 38$**	$3x + 4 = 5x - 6 \rightarrow x = 5, JM = 19 \rightarrow JK = 2(19) = 38$
7	Answer: **$x = 10$**	$2(3x + 4) = 5x + 18 \rightarrow 6x + 8 = 5x + 18 \rightarrow x = 10$
8	Answer: **$x = 30$**	$(3x + 20) + (2x + 10) = 180 \rightarrow 5x + 30 = 180 \rightarrow x = 30$
9	Answer: **$x = 14$**	$(2x + 8) + (4x - 2) = 90 \rightarrow 6x + 6 = 90 \rightarrow x = 14$
10	Answer: **$AC = 31$**	$(x + 7) + (2x - 3) = 5x - 14 \rightarrow 3x + 4 = 5x - 14 \rightarrow x = 9 \rightarrow AC = 5(9) - 14 = 31$